

# test-coverage

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## LLMOrchestrator — Symbol→Test Coverage Ledger

**Round:** 275 (deep-doc + Challenge enrichment, mirror round-220 template) **Generated:** 2026-05-19 **Mandate:** CONST-048 (full-automation coverage) + CONST-050 (no-fakes-beyond-unit-tests) + CONST-035 / Article XI §11.9 (anti-bluff)

Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17):

"all existing tests and Challenges do work in anti-bluff manner — they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

## Purpose

This ledger maps every exported symbol in the LLMOrchestrator submodule to the test or Challenge that exercises it with runtime evidence. A row without runtime evidence is a §11.4 PASS-bluff regardless of how green the green column reads.

## Convention

- **Layer:** unit (mocks allowed per CONST-050(A)), integration (real collaborators, no mocks), challenge (out-of-process bash wrapper, real subprocess + real disk + real wire).
- **Evidence:** anti-bluff artefact emitted during the test (file path, stdout substring, captured JSON). A green PASS column without an evidence column is a critical defect.
- **Mutation:** for every gate, a paired meta-test mutation (§1.1) that inverts the polarity of one invariant. The mutation MUST flip green→red.

## Coverage Ledger

| Package      | Symbol                                 | Layer      | Test / Challenge                                         |
|--------------|----------------------------------------|------------|----------------------------------------------------------|
| pkg/parser   | DefaultParser.Parse                    | unit       | pkg/parser/<br>parser_test.go::TestDefault               |
| pkg/parser   | DefaultParser.Parse (empty)            | unit       | pkg/parser/parser_test.go:                               |
| pkg/parser   | DefaultParser.Parse (5-locale prompts) | challenge  | challenges/runner/main.go:<br>llorchestrator_describe_c  |
| pkg/parser   | DefaultParser.ExtractJSON              | unit       | parser_test.go::TestExtract                              |
| pkg/parser   | DefaultParser.ExtractActions           | unit       | parser_test.go::TestExtract                              |
| pkg/parser   | DefaultParser.ExtractIssues            | unit       | parser_test.go::TestExtract                              |
| pkg/protocol | PipeMessage JSON encoding              | challenge  | challenges/runner/main.go:<br>pipe_test.go::TestPipeMess |
| pkg/protocol | MessageTypePrompt constant             | challenge  | challenges/runner/main.go:                               |
| pkg/protocol | NewFileTransport                       | challenge  | challenges/runner/main.go:<br>file_test.go::TestNewFileT |
| pkg/protocol | FileTransport.WriteToInbox             | challenge  | challenges/runner/main.go:<br>file_test.go::TestWriteToI |
| pkg/protocol | FileTransport.ReadFromInbox            | challenge  | challenges/runner/main.go:<br>file_test.go::TestReadFrom |
| pkg/protocol | FileTransport.WriteSharedFile          | unit       | file_test.go::TestSharedFi                               |
| pkg/protocol | validatePath (private, exercised)      | unit+integ | file_test.go::TestPathTrav<br>protocol_integration_test. |
| pkg/protocol | PipeTransport.Send / Recv              | unit       | pkg/protocol/pipe_test.go                                |
| pkg/i18n     | Translator interface                   | unit       | pkg/i18n/<br>translator_test.go::TestTr                  |

| Package          | Symbol                        | Layer      | Test / Challenge                                         |
|------------------|-------------------------------|------------|----------------------------------------------------------|
| pkg/i18n         | NoopTranslator.T              | challenge  | challenges/runner/main.go:                               |
| pkg/i18n         | NoopTranslator.TPlural        | challenge  | challenges/runner/main.go:variant)                       |
| pkg/i18n         | SetPkgTranslator(nil) + Pkg() | challenge  | challenges/runner/main.go:(Pkg.reset_to_noop)            |
| pkg/agent        | Agent interface               | unit+integ | pkg/agent/agent_test.go + *_agent_test.go                |
| pkg/agent        | AgentPool.Acquire             | unit       | pkg/agent/pool_test.go::Te                               |
| pkg/agent        | SimplePool round-robin        | unit       | pkg/agent/simple_pool_test                               |
| pkg/agent        | HealthMonitor circuit-breaker | unit       | pkg/agent/health_test.go::TestHealth                     |
| pkg/agent        | ClaudeCodeAgent.Send          | unit       | pkg/agent/claudecode_agent                               |
| pkg/agent        | GeminiAgent.Send              | unit       | pkg/agent/gemini_agent_tes                               |
| pkg/agent        | JunieAgent.Send               | unit       | pkg/agent/junie_agent_test                               |
| pkg/agent        | OpenCodeAgent.Send            | unit       | pkg/agent/opencode_agent_t                               |
| pkg/agent        | QwenCodeAgent.Send            | unit       | pkg/agent/qwencode_agent_t                               |
| pkg/agent        | MultiPool.Add / Pick          | unit       | pkg/agent/mock_pool_test.g                               |
| pkg/adapter      | BaseAdapter lifecycle         | unit+integ | pkg/adapter/adapter_test.g<br>adapter_integration_test.g |
| pkg/adapter      | OpenCodeHeadless.Run          | unit       | pkg/adapter/opencode_headl                               |
| pkg/config       | FromEnv                       | unit       | pkg/config/config_test.go:                               |
| cmd/orchestrator | main entry                    | integ      | automation_test.go::Test0r                               |
| Chaos            | failure injection across pool | challenge  | challenges/scripts/chaos_failure_injection_ch            |
| DDoS             | health-endpoint flood         | challenge  | challenges/scripts/ddos_health_flood_challeng            |
| Scaling          | horizontal pool growth        | challenge  | challenges/scripts/scaling_horizontal_challen            |
| Stress           | sustained-load mix            | challenge  | challenges/scripts/stress_sustained_load_chal            |
| UI               | terminal interaction          | challenge  | challenges/scripts/ui_terminal_interaction_ch            |
| UX               | end-to-end flow               |            |                                                          |

| Package | Symbol | Layer          | Test / Challenge                                  |
|---------|--------|----------------|---------------------------------------------------|
|         |        | chal-<br>lenge | challenges/scripts/<br>ux_end_to_end_flow_challen |

# Invariant Floor (per CONST-048)

For every row above, six invariants are asserted:

- 1. **Anti-bluff posture** — captured runtime evidence per §11.4.
- 2. **Proof of working capability** — end-to-end on the documented topology.
- 3. **Implementation matches docs** — README + USER\_GUIDE + this ledger reflect actual API.
- 4. **No open issues** — docs/Issues.md empty for this row or row marked OPERATOR-BLOCKED per §11.4.21.
- 5. **Documentation in sync** — .md + .html + .pdf mtimes lockstep per §11.4.12 / §11.4.53.
- 6. **Four-layer test floor** — pre-build + post-build + runtime + paired mutation.

# How to Re-validate

```
# Round-275 Challenge (this round's deliverable)
cd dependencies/HelixDevelopment/LLMOrchestrator
bash challenges/llmorchestrator_describe_challenge.sh normal
# → exit 0
bash challenges/llmorchestrator_describe_challenge.sh mutate
# → exit 99

# Full unit + integration with race detector
go test -race -count=1 ./...

=====
```

# LLMOrchestrator Test Coverage Ledger (round-291, vasic-digital)

This ledger maps every exported symbol in pkg/{adapter,agent,config,parser,protocol} to the test (unit, integration, fuzz, security, stress) that exercises it and to the round-291 Challenge-runner invariant (challenges/runner/main.go) that re-validates the user-visible behaviour at end-user altitude.

It is the visible audit surface that closes the §11.4 anti-bluff covenant: **every PASS in this submodule MUST correspond to a row below where the Challenge column is non-empty** — otherwise the PASS is a metadata-only check (CONST-035).

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## How to read this table

| Column                     | Meaning                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symbol</b>              | Exported Go identifier (Package.Func / Package.Type.Method).                                                                                                                                                         |
| <b>Source</b>              | File and line where the symbol is declared.                                                                                                                                                                          |
| <b>Test(s)</b>             | *_test.go files that exercise the symbol with assertions.                                                                                                                                                            |
| <b>Challenge invariant</b> | Numbered invariant in challenges/runner/main.go that re-proves the symbol works end-to-end across the 5-locale fixture matrix. Empty = covered by tests only; non-empty = also re-validated by the round-291 runner. |
| <b>Bluff risk closed</b>   | Concrete user-visible failure mode the assertion catches.                                                                                                                                                            |

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### pkg/parser

| Symbol              | Source       | Test(s)                                                            | Challenge          |
|---------------------|--------------|--------------------------------------------------------------------|--------------------|
| NewParser           | parser.go:71 | parser_test.go (all TestParser_*)                                  | (1)                |
| DefaultParser.Parse | parser.go:76 | parser_test.go (TestParser_Parse_*);<br>parser_fuzz_test.go (fuzz) | (1)<br>pars<br>(2) |

| Symbol                                                                          | Source          | Test(s)                                                                  | Cha         |
|---------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------|-------------|
| DefaultParser.ExtractJSON                                                       | parser.go:112   | parser_test.go<br>(TestParser_ExtractJSON_*);<br>parser_security_test.go | (1)         |
| DefaultParser.ExtractActions                                                    | parser.go:150   | parser_test.go<br>(TestParser_ExtractActions_*)                          | (1)         |
| DefaultParser.ExtractIssues                                                     | parser.go:200   | parser_test.go<br>(TestParser_ExtractIssues_*)                           | (cov<br>Par |
| ErrEmptyInput /<br>ErrNoJSONFound /<br>ErrMalformedJSON /<br>ErrResponseTooLong | parser.go:17-25 | parser_test.go;<br>parser_security_test.go<br>(length-DoS)               | (2)<br>(spe |
| MaxResponseLength                                                               | parser.go:28    | parser_security_test.go                                                  | —           |

## pkg/protocol

| Symbol               | Source        | Test(s)                                                                        | Cl        |
|----------------------|---------------|--------------------------------------------------------------------------------|-----------|
| PipeMessage (struct) | message.go:25 | message_test.go;<br>pipe_test.go (round-trip);<br>protocol_integration_test.go | (3)<br>pr |

| Symbol                                          | Source           | Test(s)                                       | Cl       |
|-------------------------------------------------|------------------|-----------------------------------------------|----------|
| MessageType constants                           | message.go:13-22 | message_test.go                               | (3       |
| FileMessage / FileAttachment                    | message.go:45-60 | file_test.go;<br>protocol_integration_test.go | (4<br>pr |
| NewFileTransport                                | file.go:34       | file_test.go<br>(TestFileTransport_*)         | (4       |
| FileTransport.WriteToInbox /<br>ReadFromInbox   | file.go:56,66    | file_test.go;<br>protocol_integration_test.go | (4       |
| FileTransport.WriteToOutbox /<br>ReadFromOutbox | file.go:61,71    | file_test.go                                  | —        |
| FileTransport.WriteSharedFile                   | file.go:76       | file_test.go (path-tra-<br>versal)            | —        |
| ErrSessionDirNotExist /<br>ErrPathTraversal     | file.go:17-22    | file_test.go (security)                       | —        |
| PipeTransport.*                                 | pipe.go          | pipe_test.go                                  | —        |

| Symbol | Source | Test(s) | Challenge inv |
|--------|--------|---------|---------------|
|        |        |         |               |

pkg/agent

| Symbol                                                                                                                                                     | Source          | Test(s)                                               | Challenge inv                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------------------------------------|------------------------------------|
| Agent (interface)                                                                                                                                          | agent.go:12     | (every adapter<br>_test.go imple-<br>ments & asserts) | (5) (via Register)                 |
| Response / StreamChunk /<br>Attachment /<br>AgentCapabilities /<br>AgentRequirements /<br>HealthStatus / Action /<br>ParsedResponse / Issue /<br>ModelInfo | agent.go:42-127 | (used throughout<br>the test surface)                 | (5)                                |
| NewPool                                                                                                                                                    | pool.go:54      | pool_test.go;<br>pool_stress_test.go                  | (5) agent.Pool.                    |
| Pool.Register                                                                                                                                              | pool.go:63      | pool_test.go                                          | (5) agent.Pool.                    |
| Pool.Acquire                                                                                                                                               | pool.go:87      | pool_test.go;<br>pool_stress_test.go<br>(concurrency) | (5) agent.Pool.<br>agent.Pool.Acqu |



| Symbol                                                  | Source          | Test(s)                         | Challenge inv               |
|---------------------------------------------------------|-----------------|---------------------------------|-----------------------------|
| Pool.Release                                            | pool.go:169     | pool_test.go                    | (5) agent.Pool.             |
| Pool.Available /<br>Pool.HealthCheck /<br>Pool.Shutdown | pool.go:180-216 | pool_test.go;<br>health_test.go | —                           |
| MultiPool.*                                             | multi_pool.go   | pool_test.go (sub-<br>set)      | —                           |
| CircuitBreaker.* (health)                               | health.go       | health_test.go                  | —                           |
| ErrAgentAlreadyRegistered /<br>ErrNoAvailableAgent /    | pool.go:13-21   | pool_test.go                    | (5) (via Register<br>paths) |

| Symbol                                | Source | Test(s) | Challenge inv |
|---------------------------------------|--------|---------|---------------|
| ErrPoolShutdown /<br>ErrAgentNotFound |        |         |               |

pkg/adapter

| Symbol                                                    | Source            | Test(s)                                    |
|-----------------------------------------------------------|-------------------|--------------------------------------------|
| BaseAdapter                                               | base.go           | adapter_test.go;<br>adapter_integration_te |
| NewOpenCodeAgent /<br>OpenCodeAgent.parseOpenCodeResponse | opencode.go:20,53 | adapter_test.go;<br>opencode_headless_test |
| NewClaudeCodeAgent                                        | claudecode.go     | adapter_test.go                            |
| NewGeminiAgent                                            | gemini.go         | adapter_test.go                            |

| Symbol                | Source               | Test(s)                |
|-----------------------|----------------------|------------------------|
| NewJunieAgent         | junie.go             | adapter_test.go        |
| NewQwenCodeAgent      | qwencode.go          | adapter_test.go        |
| OpenCodeHeadlessAgent | opencode_headless.go | opencode_headless_test |
| AdapterConfig         | base.go              | adapter_test.go        |

pkg/config

| Symbol                                                | Source    | Test(s)        | Challenge invariant                          | Bluff risk closed                                                  |
|-------------------------------------------------------|-----------|----------------|----------------------------------------------|--------------------------------------------------------------------|
| Config / LoadFromEnv / LoadFromEnvironment / Validate | config.go | config_test.go | —<br>(covered by tests; runner uses no .env) | .env parser silently skipping required fields → orchestrator boots |

| Symbol                       | Source    | Test(s)        | Challenge invariant | Bluff risk closed                                                               |
|------------------------------|-----------|----------------|---------------------|---------------------------------------------------------------------------------|
| AgentBinaryPath / SessionDir | config.go | config_test.go | —                   | with no agents and no warning. Path-traversal in user-supplied SessionDir base. |

## Test-type coverage summary

| Type               | Where                                                                              | Anti-bluff guarantee                                                   |
|--------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Unit</b>        | pkg/*/*_test.go                                                                    | Direct API exercise; mocks ALLOWED here only per CONST-050(A).         |
| <b>Integration</b> | pkg/adapter/adapter_integration_test.go; pkg/protocol/protocol_integration_test.go | Real os/exec / real os.MkdirTemp / real JSON; no fakes (CONST-050(A)). |
| <b>Fuzz</b>        | pkg/parser/parser_fuzz_test.go                                                     | go test -fuzz=driven; surfaces panics on adversarial input.            |
| <b>Security</b>    | pkg/parser/parser_security_test.go                                                 | Length-DoS, traversal, command-injection patterns.                     |
| <b>Stress</b>      | pkg/agent/pool_stress_test.go                                                      |                                                                        |

| Type                              | Where                                                                            | Anti-bluff guarantee                                                                         |
|-----------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                                   |                                                                                  | Concurrent Acquire/Release under contention; race detector mandatory.                        |
| <b>Challenge (round-291)</b>      | challenges/runner/main.go + challenges/<br>llmorchestrator_describe_challenge.sh | 5 invariants × 5 locales = 22 PASS rows; paired \$1.1 mutation.                              |
| <b>Challenge (legacy)</b>         | challenges/scripts/*.sh                                                          | UI / UX / DDoS / chaos / scaling / stress / no-suspend / host-hardening (legacy bash suite). |
| <b>Performance / benchmarking</b> | Not yet wired — tracked as future work (CONST-048).                              | —                                                                                            |

## Anti-bluff guarantees

1. **No metadata-only PASS.** Every Challenge-column entry in the table above corresponds to a runtime invariant in challenges/runner/main.go that compares observed values against fixture-pinned expected values byte-for-byte.
2. **5-locale matrix.** Every protocol-level invariant runs across en, sr, ja, es, de — a single English-only PASS would not survive CONST-046 (no-hardcoded-content) review.
3. **Paired mutation.** challenges/llmorchestrator\_describe\_challenge.sh mutate flips one invariant and asserts the runner exits non-zero. Mutation-undetected = runner is a bluff gate.
4. **Real I/O.** The runner uses os.MkdirTemp + os.WriteFile (via protocol.FileTransport) — never an in-memory shim. The Pool exercises Register/Acquire/Release on a real adapter.NewOpenCodeAgent instance.
5. **No mocks past the unit-test boundary.** Per CONST-050(A), the runner's path is real types, real I/O, real concurrency. pkg/agent/multi\_pool.go etc. are excluded from runner only because their behaviour is a strict superset of Pool already covered.

## How to run

```
cd <repo>/dependencies/vasic-digital/LLMOrchestrator
```

```
# Normal mode – every invariant must PASS, exit 0
```

```
bash challenges/llmorchestrator_describe_challenge.sh normal
```

```
# Mutation mode – one invariant flipped; runner MUST exit non-  
zero,
```

```
# wrapper rewrites to 99 (paired-mutation success)
```

```
bash challenges/llmorchestrator_describe_challenge.sh mutate
```

Both invocations are wired into the round-291 evidence capture.

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## Verbatim 2026-05-19 operator mandate (preserved per CONST-049 §11.4.17)

"all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!"

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